



BY AMETECS

SR-09

Five-Session Engineering Report

Sessions of 11–14 August 2026 · Product, Central and Agent

Version 1.0

15 August 2026

Prepared for: Ejaz Hussain, MD

AMETECS INDIA PRIVATE LIMITED · HYDERABAD

Contents

01	Executive Summary
02	Purpose, Scope and Method
03	Session S1 — 11 August 2026
04	Session S2 — 12 August 2026
05	Session S3 — 13 August 2026 (day)
06	Session S4 — 13 August 2026 (evening)
07	Session S5 — 14 August 2026
08	Commit Register
09	Verified Repository State — 15 August 2026
10	Repository Hygiene Findings
11	What Changed
12	What Is Pending
13	Decisions Needed From You
14	Lessons Recorded

1. Executive Summary

Read this first to understand where SmartEPT stands after the workstation change, what the last five sessions delivered, and what is waiting on you.

Between 11 and 14 August 2026, five working sessions delivered thirty-eight commits across the two SmartEPT repositories. The work falls into four themes: commercial pricing became progressive and custom-quotable; licensing moved from a single-tenant model to per-tenant with real seat enforcement; a long chain of live faults was root-caused and closed; and the first on-premise client package was built.

Three findings deserve your attention before anything else.

Everything is committed and pushed. Verified against both repositories on 15 August. Central sits at `2cf456c`, Product at `5d62cb9`, both identical to their remotes, both with clean working trees. The project memory said otherwise; it was written mid-session and was wrong. That is corrected.

None of it is live. The 14 August licence work — seat caps, expiry blocking, trial supersession — exists in the repository but has not been migrated or deployed. Until it is, no client is affected by any of it.

The agent your clients download is six versions old. They are receiving version 0.5.0, built 20 July. Version 0.11.0 was built on 25 July and never published. The agent has no auto-updater, so every correction reaches a client only through a manual reinstall. This is the single item on the outstanding list with direct customer impact today.

Two decisions are waiting on you and are set out in section 13. One concerns whether the super admin should be exempt from licence blocking on the shared cloud server. The other concerns what happens to Ametecs' own company when the per-tenant licensing migration runs on live and the installation licence slot empties.

2. Purpose, Scope and Method

This report reconstructs the five working sessions of 11, 12, 13 (day), 13 (evening) and 14 August 2026, and states the verified condition of both source repositories as at 15 August 2026.

It was prepared after the development workstation was replaced. The new machine carries the same working trees and the same project memory as the old one, but the running knowledge of what had been finished, what had been pushed and what was still waiting did not travel with it. This document restores that.

Two categories of statement appear below and they are deliberately kept apart. **Recorded** means the session that did the work wrote it down. **Verified** means it was checked against the

repository on 15 August 2026 and found to be true. Where the two disagree, the disagreement is stated rather than quietly resolved — section 12 contains one such case, and it is the reason section 1 opens with the push status.

The verification method, for the record: the Git index of each repository was read directly and every tracked entry compared against the file on disk for size and modification time, and local branch references compared against remote-tracking references. No shell was available on the workstation, so this was done by reading the repository's own metadata over the file bridge.

3. Session S1 — 11 August 2026

3.1 Progressive lifetime pricing

On-Premise lifetime pricing changed from flat per-band prices to milestone pricing with straight-line interpolation. A band's stated price is now the price **at that band's maximum user count**. A customer between two milestones pays the interpolated figure rather than the higher band price.

Your configured bands are: 5–15 at ₹15,000 · 16–30 at ₹25,000 · 31–50 at ₹40,000 · 51–100 at ₹60,000 · 101–150 at ₹75,000 · 151–200 at ₹90,000.

Worked example — a 41-user licence. Forty-one users sits between the 30-user milestone (₹25,000) and the 50-user milestone (₹40,000). The gap is ₹15,000 across 20 users, so ₹750 per user. Eleven users past the milestone gives ₹25,000 + (11 × ₹750) = **₹33,250**. Under the old flat model the same customer paid the full ₹40,000. AMC at eighteen per cent is charged on the interpolated figure: 18% of ₹33,250 = **₹5,985 per year**, not ₹7,200.

Prices inside a band are therefore lower than before by design. If the economics need adjusting, the milestones themselves are edited in Super Admin — the interpolation takes care of everything between them.

Three protections were built in. The first band is a flat minimum package and is never reduced below its price. Below the first band's minimum the system refuses the order with a message naming the minimum capacity. An open-ended top band means Custom Quote with the price stored as NULL, and `BillingService::createOrder` throws rather than ever writing a zero-rupee lifetime licence.

The interpolation is mirrored in four places that must agree: the pricing engine, the `/buy` page, the client authentication screen, and the landing page widget. The landing page is a static file rebuilt from the CMS, so `php artisan landing:import` must be run or a later CMS publish will silently revert the fix.

3.2 The seeded login defect

No seeded login had ever worked. The seeder applied `Hash::make` to a field that already carried Laravel's `hashed` cast, so every password it wrote was hashed twice. The defect is invisible in

testing because it presents as an ordinary "credentials do not match" rejection rather than as an error.

Fixing the code does not repair rows already written. Existing accounts must be unlocked through tinker using `forceFill()->save()` — never `where()->update()`, which bypasses the cast and reintroduces the problem. Every seeded password should be treated as having been effectively unset, and changed accordingly.

3.3 Client-wise SMTP and password recovery

Mail now resolves in a defined order: the company's own SMTP settings first, then the global relay, then the environment file. Console forgot-password by email OTP was added for all roles. On a fresh installation with no mail configured, OTP codes land in the application log rather than an inbox, which is the fallback route when an administrator is locked out.

3.4 Live server triage

Four faults on the production server were found and closed.

The licence signing keys existed only on the development machine, having been deliberately excluded from version control. On your explicit instruction they were committed to the repository.

The repository must therefore remain private permanently — this is recorded in the licensing standard as an owner override.

The hosted console link was unset, so cloud provisioning had nowhere to point. The Interakt WhatsApp key was supplied and configured.

The hourly dunning job had been failing twice every hour. It writes a tenant status of `purged`, but the column was an ENUM that did not include that value, so the write failed and the job died.

This was the third enum failure of its kind. Two more followed within the sessions covered by this report; section 14 collects them.

3.5 Commercial and administrative

Razorpay verified smartept.com on 11 August. The remediation that led to verification — subscription framing rather than rental, consent-framed copy, and the legal pages — was applied and committed.

Super-admin-only deletion was added for quotations and unpaid orders, permitted only where no ledger entry and no licence exist, and for invoices with a financial-year gap warning. **Money-bearing orders are never deletable.** The coupon box on the buy page is now hidden unless a coupon is actually live. A sweep removed trademark and registered symbols from the product, per the standing rule that nothing is registered yet and only the copyright line is permitted.

4. Session S2 — 12 August 2026

4.1 Why the super admin saw another company's licence

You reported that the super admin console was displaying Khan Incorporation's licence. The cause was structural rather than cosmetic.

The `installation_licenses` table held a **single row per installation**. That is correct for a client hosting SmartEPT on their own server. It is wrong for the cloud installation, which hosts several companies behind one application. Khan's key occupied the only slot, so the entire installation was running on Khan's twenty-five seats, and all installation storage was being reported to Central under Khan's key.

The model was corrected and the concept fixed in your words — *"central is the super admin, product have multiple tenants"*:

- **Central** is the business super admin and the sole commercial truth: orders, billing, licences.
- The **Product super admin** is a host operator. It performs cross-tenant operations and owns no licence of its own.
- **Each cloud tenant carries its own licence row**. The installation slot remains, but now serves only companies that are not cloud-hosted.

A Tenants screen was added at the top of the navigation for host operations, showing per company: licence health, devices bound against seats, users, employees, storage used against quota with a percentage, last heartbeat, and provisioning status.

One consequence needs a decision before the live migration runs. When the migration moves the installation key to Khan, the installation slot empties. Companies that are not cloud-hosted — including Ametecs' own — then begin a fresh seven-day evaluation, after which their agents block. This is covered in section 13.

4.2 Licence change flows

A survey of the code against your question — *"upgrade, downgrade, seats and billing not found anywhere; how do I reissue a licence file"* — found most of it genuinely absent.

Cloud pro-rata upgrade existed, but only in the client portal, behind a card that was hidden. Administrators had no billed upgrade at all, only the ability to silently edit a device limit without raising an invoice. Downgrade had no mechanism whatsoever — "reductions at renewal" was a comment in the source and nothing more, and renewals always billed the current size. Perpetual upgrades were hard-blocked. Licence file reissue was administrator-only.

Four policies were settled with you and built in a single commit.

Perpetual licences upgrade but never downgrade, and are never refunded. Once purchased, the capacity is owned.

A perpetual upgrade is priced as the difference between interpolated lifetime prices.

Worked example, the one used to validate the build: a customer on a 15-user perpetual licence upgrading to 30 users. The 15-user price is ₹15,000 and the 30-user price is ₹25,000, so the upgrade order is ₹10,000 plus eighteen per cent GST of ₹1,800 — **₹11,800 payable**. It is raised as a line of type `upgrade` on the same GST invoice series, so the financial-year numbering stays continuous. AMC re-bases automatically: the next AMC bill is calculated from 30 users, not 15.

Cloud downgrades apply at renewal only, and never below the count of active devices.

Both the client and an administrator can schedule one; mid-period reductions remain impossible. The scheduled figure is billed at the next renewal and then cleared.

Clients may re-download their own licence file, but only against the fingerprint already stored. Moving a licence to a different machine remains an administrative action through Shift Machine. This is a security boundary, not a convenience limit.

4.3 The test suite had been broken since 19 July

A test run returning "5 failed, 0 assertions" led to three stacked causes, fixed from the bottom up.

A July migration used raw MySQL `ALTER TABLE` with no driver guard, so it could not run under the test database. A machine-level `DB_CONNECTION` environment variable on the workstation was overriding PHPUnit's own configuration, which does not take precedence over operating-system variables unless forced. Underneath both, `config/database.php` hard-coded the default connection to MySQL — a fix made on 19 July against a rogue environment variable — so PHPUnit's SQLite setting could never apply at all.

The consequence is worth stating plainly. **Every Product feature test had been failing silently since 19 July**, and the suite had not been run in the interval, so nobody saw it. Once revived, the tests immediately caught a genuine defect: a 500 error in `ProvisionController` for any caller omitting an optional field. Central always sends that field, so production never saw it — but a client's own integration would have.

4.4 The agent's HTTP 500

A device registration was failing with a server error. The log named a duplicate-key violation on the device identifier. The device had registered earlier under a **different company**, and the lookup was subject to company scoping, so it could not see the existing row and attempted an insert.

The symptom pointed the wrong way entirely — "the main console works, the branded URLs do not" — which was a coincidence of which accounts were used, not a URL problem. The fix removes scoping from lookups on globally-unique columns, re-points the row to the new company, and audits the transfer.

4.5 Search visibility and documentation

The sitemap was emitting insecure URLs because it was built from a configuration value that had drifted; it now forces the secure scheme regardless. Canonical tags were added to seven public pages, and workflow pages marked no-index without blocking crawlers outright. The GA4 tag was absent because publishing rendered from an empty CMS table; publishing now repairs itself first, and tracking tags extend to the legal pages and the buy page.

Documents SR-04 through SR-07, the MR-01 consolidated report, the MR-02 120-case test suite, the OPS-01 recovery guide, and the redesigned six-page WhatsApp campaign report with its sixty-second status video were produced in this session.

5. Session S3 — 13 August 2026 (day)

Six faults were root-caused and closed. They are listed together because they formed a chain: each one was masking the next.

OTP codes were being returned in API responses. A test-mode convenience was gated on debug mode, and production was running with debug mode enabled. Both the leak and the setting were corrected.

Cloud provisioning discarded the console's temporary password instead of sending it, so every cloud client was locked out of their own console at the moment of provisioning. It is now emailed.

Shared PHP workers were leaking the database name between the two applications. Both applications run under the same web server, and `putenv` writes into the process environment, which the next request on that worker inherits. This is the true root cause behind the July wrong-database incidents, the agent and screenshot faults attributed to them, and the folklore about restarting Laragon that grew up around them. It also explains a licence fingerprint that had been bound during the affected period. Disabling `putenv` in both applications closes it at the root.

The agent's 405 error came from a redirect that Windows followed as a GET request. **Licence validation failed for the same reason** on a different redirect, from insecure to secure. Both are now handled strictly rather than by redirect.

Console installations had no mail configuration at all, so they now inherit Central's relay at provisioning and adopt it as their global relay only if none is set.

5.1 The on-premise client package

Built in the same session. The SmartPRS2 licence logic was ported across unchanged, as you asked — the machine fingerprint is cached and persisted so a flaky reading cannot invalidate a licence, floating licence files with no fingerprint are refused outright, the enforcement gate fails

soft so that any internal error lets traffic through while real verdicts still block, and an activation page sits before login so a client can upload their licence file without signing in first.

The installer kit covers three platforms: Windows through Laragon, Linux with a systemd service, and macOS with a launch agent, all sharing one helper. The package rebuild script produces a versioned archive that the portal serves automatically.

6. Session S4 — 13 August 2026 (evening)

6.1 Custom pricing

Administrators may now set a custom price, a custom setup charge and a custom AMC on orders, quotations and prospect quotes, at any user count. The custom setup charge replaces the calculated fee and is forced onto the order even where setup has already been paid; the custom AMC is added as its own line. Access is limited to super admins and to staff holding billing rights.

6.2 The request queue

Beyond the last priced band — above 200 users on your live configuration — the buy page no longer quotes a price. It captures a **request**: the customer's details, their requirement and their notes, with **no price shown at any point**. The request lands in the orders queue badged by origin, so you can see at a glance whether it came from a client or from an administrator. From there it is edited, priced, and converted in place, which sends the quotation with print and payment links.

Requests take an order number but **no quotation number until conversion**. This keeps the financial-year quotation series unbroken — an abandoned request never consumes a number.

6.3 Decisions recorded

Perpetual terms remain lifetime. The "minimum fifteen years" line is a commercial talking point for conversations with clients and must not become an expiry date in the system. No change was made.

Requests are fully editable; converted numbered quotations are not. The existing rule stands for anything numbered — delete and re-create, never edit. A request is editable precisely because it is pre-number and pre-money.

6.4 Test results

The session ended with the Central suite at **69 passed, 391 assertions, in 14.3 seconds**. Two rounds of failure preceded that. The first produced six failures: four from SQLite enforcing enum constraints that the MySQL-only migrations had never addressed, and two from a security change made earlier the same day that removed OTPs from API responses, which the tests had

been reading. The second produced one, and it was a genuine defect — perpetual quotes were adding a setup fee even when setup had been explicitly declined, both from the administrator's "Include setup: No" and from the customer unticking the box. The preview and the charge disagreed. It is fixed.

7. Session S5 — 14 August 2026

Run by **Altat**, working from four findings you reported.

7.1 Trial licences stayed active after payment

A customer who subscribed kept an active trial alongside the paid licence. Licences now carry a `superseded` status, and issuing a paid licence closes the trial automatically — on first purchase, and after renewals and upgrades. Existing clients already in that state are corrected by a backfill in the migration.

The review then closed every route by which a superseded trial could come back to life: extending a trial no longer touches one, resuming is refused server-side while a paid licence is live, the interface shows "closed by paid licence" instead of a Reactivate button, and the purge sweep includes them.

7.2 No action beside status

The client card showed licence status but offered nothing to do about it. Suspend, Reactivate and Revoke now sit beside it. Revoke was also surfaced in the licences list, where the underlying action already existed but had no button.

7.3 Central showed zero of twenty-five seats for a live client

Consoles never reported their counts. They now send user, employee and device counts on the daily check-in. Central stores them and shows a Users column in both the licences list and the client card, marked red when over the limit. The counts are optional in the payload, so consoles running older builds continue to validate normally.

7.4 Seat count was a label, not a rule

The number on the licence was displayed but never enforced. A new seat service is now the single authority on whether a seat is free.

One seat means one monitored person on one PC. Active employees and employee-role logins consume seats; administrator, HR and manager logins are free.

An adversarial review of the first implementation found six bypasses, and **every one of them was an update path** rather than a creation path: rebinding a device, changing a role to employee, re-enabling a disabled employee login, moving a relieved employee back to active,

restoring an archived employee, and a bulk import whose cap was dead code because a closure had not captured the variables it needed. All are now guarded.

Counting is deliberately local. The device activation call fails open when Central is unreachable, which is precisely how an offline server could otherwise bind machines indefinitely. The cap is **not retrospective**: clients already over the count keep everyone they have, they simply cannot add more.

7.5 Expired licences did not stop the client working

Enforcement had been applied only to agent routes. It now covers the whole authenticated API, with a deliberate allow-list. Anyone may still fetch their own identity, log out, refresh a token and change a password. A super admin or company administrator may additionally reach the licence screens, licence import, diagnostics and logs — that is the rescue route, chosen as "hard block, administrator-only rescue".

Non-administrative sign-in is refused when the licence is dead, and the interface shows a single full-screen licence wall rather than every screen failing separately with its own error.

7.6 The IIS incident

A new client — the first hosted on IIS rather than Laragon — could open every screen and create records, but **every Edit and Save returned HTTP 405**, and Delete returned 405 or 404.

IIS ships with WebDAV enabled by default. WebDAV registers a handler for PUT, DELETE and related methods, and answers 405 **before** URL Rewrite or PHP ever see the request. GET and POST pass through untouched. That is exactly the reported symptom: everything works except saving an edit.

Nothing was wrong with SmartEPT. The routes exist as PUT, the console sends a real PUT, the HTTPS middleware returns a refusal rather than a redirect, and no route cache ships in the build.

The fix removes both the WebDAV module and its handler in the site configuration. **Both lines are required** — removing only one leaves the behaviour unchanged. The trailing slash redirect was also restricted to GET and HEAD, because a redirect on a PUT is retried by the browser as a GET, producing another 405. The installation guide gained the step, the one-line command that proves the diagnosis in a single request, and a warning about the site root.

Two related observations were made at that client. The package had been unzipped to a location the guide says not to use, which is harmless only if the site's physical path points at the public subfolder — if the environment file downloads over the web, it does not, and that must be corrected before go-live. And the package rebuild script does not exclude one cache directory, so if configuration caching is run before a package is built, development configuration including database credentials ships to the client. This was raised and has not been fixed.

7.7 The agent is six versions behind

Clients are downloading **version 0.5.0, built 20 July** — the newest build published to Central's downloads. **Version 0.11.0 was built on 25 July and never published.**

The branded-slug fix was committed and the version bumped to 0.12.0, but the build and the publish, to both the local and the live download sources, have not been done.

The agent has **no auto-updater**. Every update reaches a client only through a manual reinstall, and rollout can be confirmed only by checking reported application versions in the console's device list. The full procedure is in the RU-01 runbook produced the same session.

7.8 Documentation

Two documents were produced: **RU-01 Release and Update Runbook**, fourteen pages, the lasting reference for how any build reaches a client; and **SR-08 Session Report**, ten pages, covering that session. Both carry an internal part and a client-facing part that can be sent on its own.

8. Commit Register

Central — smartept-central

Date	Commits
11 Aug	e6f5839 d8279a5 2d9d8a6 385836a 73888fc 72c0059 a03f228 6360796
12 Aug	3ce729f 706b642 44f71d2 2fa3fab 76ddf09 face8e7 a055ed3 558417c b6be8c8
13 Aug	d768719 9a2e3f5 e229a7b 2c0aa61 19a1b09 00d929a ef29cd4 34a3e77 16640d1
14 Aug	11b2d17 3ad8a08 2cf456c

Product — smartept

Date	Commits
11 Aug	22ed12e
12 Aug	1f6be35 1fcb7f7 01174cc aef81df 408785c 2ca7dec 389b8a0 979ccba 1c12fb0 1c4cdcb
13 Aug	0b2e4e5 9d1bae5 ec1f78f 63fc4bd f5e2d2b

Date	Commits
14 Aug	9a1854e 436f0ef 5d62cb9

Thirty-eight commits in total across the five sessions.

9. Verified Repository State — 15 August 2026

Repository	Local main	Remote	Working tree
smartept-central	2cf456c	identical	clean
smartept	5d62cb9	identical	one untracked file

The last push to Central was made on 14 August at 19:53, and to Product at 19:54. All 283 tracked files in Central and all 348 in Product match the index in both size and modification time. Nothing on disk is newer than the index in either repository. The single untracked file is a build log in the Product deployment folder.

This corrects the project memory. The 14 August session note records that nothing had been pushed, because it was written while the session was still in progress. Both repositories were in fact pushed the same evening. The memory has been amended so that the next session does not chase work already done.

10. Repository Hygiene Findings

Verification surfaced three problems unrelated to the sessions' subject matter, recorded here so they are not lost.

Central is carrying 5.4 MB of development residue in version control — twenty-two files. Eleven are timestamped backups of the landing page, about 4.3 MB between them. Four are artefacts created by the file bridge. The remainder are backup copies of a template, a renderer and a seeder, plus two test output files.

The reason they keep accumulating is that **Central's ignore file has no rule for bridge artefacts, although the Product repository does.** Adding that one rule stops the recurrence. Removing the files already tracked is a separate and optional decision, and one I would not take without asking.

The Product repository tracks four archive files totalling 65 MB in a cloud-sync folder. These are almost certainly not meant to be under version control. Removing them from the working tree does not shrink the repository, because the history retains them; that would need history rewriting, which is not proposed here.

The client packaging script omits one cache directory from its exclusions, described in section 7.6. Raised 14 August, still open.

11. What Changed

Area	Before 11 August	After 14 August
Lifetime pricing	Flat price per band	Milestone prices, interpolated between
Above top band	No path	Request queue, priced by you, converted in place
Perpetual upgrade	Blocked	Priced as interpolated difference, GST invoiced
Cloud downgrade	No mechanism	Scheduled at renewal, floored at active devices
Licence per install	One row for all companies	One row per cloud tenant
Seat count	Displayed only	Enforced on every create and update path
Expired licence	Agent routes only	Whole API, with administrator rescue
Trial after payment	Stayed active	Automatically superseded
Console seat reporting	None	Daily counts, shown and flagged
On-premise install	Not packaged	Three-platform kit with activation page
Product test suite	Silently failing since 19 July	Running

12. What Is Pending

#	Action	State
1	Restart Laragon in both applications to clear the opcode cache	Not done
2	Run the Central migration, including the trial backfill	Not confirmed
3	Run the test suite in both repositories	Disputed — see below
4	Push both repositories	Done — verified 15 August
5	Deploy live: pull and migrate Central, pull Product	Not confirmed
6	Re-check the affected client's trial status after deployment	Blocked on 5
7	Build and publish agent 0.12.0, locally and to the live source	Not done
8	Correct the mislabelled perpetual licence and remove the wrong order	Not done

#	Action	State
9	Rebuild the client package; test one Windows and one Linux install	Not done
10	Apply code protection before wide distribution	Not done
11	Add the bridge-artefact ignore rule to Central	Newly raised
12	Fix the packaging script's cache exclusion	Raised 14 Aug, open

Items 1, 2, 3 and 5 are the deployment chain for the 14 August licence work. Until they are complete, seat enforcement and expiry blocking exist in the repository but are not in effect for any client.

On item 3, the disagreement noted in section 2: the 14 August session recorded that the test suite was never run, because that session had no PHP available. However, Central's phpunit result cache was written on 14 August at 18:05, which means a run happened. The cache is excluded from version control, so its contents cannot be inspected and no conclusion should be drawn about what passed. **Re-run rather than assume.**

On item 8, for clarity: the "pro-rata on a perpetual licence" you reported is a data problem, not a code problem. That licence is stored as a subscription with an expiry date in 2029. The correction is to edit the licence to kind Perpetual with a blank expiry, delete the incorrect unpaid upgrade order, and redo the upgrade.

13. Decisions Needed From You

D1 — The super admin exemption from licence blocking. The super admin is exempt from the console-wide block only on a shared cloud installation that carries no licence key of its own. The reasoning: on the shared box the super admin is an Ametecs operator, and blocking that account would remove our own access to the tenant list; on a client's own server the super admin is the client's owner, and receives only the rescue routes. This was chosen during the 14 August session and flagged for your approval. **It has not been approved.**

D2 — The empty installation slot after migration. When the per-tenant licensing migration runs on live, the installation-level slot empties as the key moves to its rightful company. Companies that are not cloud-hosted — including Ametecs' own — then begin a seven-day evaluation, after which their agents block. This needs either a licence issued deliberately, or a conscious acceptance, **before** the migration runs.

D3 — Repository cleanup (optional, low urgency). Whether to remove the 22 tracked residue files from Central and the four archives from Product. The ignore rule alone stops new ones accumulating; removing the existing files is a separate change to a repository that is currently clean, and I would not do it without your word.

14. Lessons Recorded

Enum columns have now failed five times. Three of the five fell within these sessions. Every recurrence has the same shape: code writes a value the column does not permit, and the failure surfaces far away from the change that caused it. New status values belong in VARCHAR columns, not enums.

A migration containing raw SQL must carry a driver guard. Two separate incidents trace back to this. Without the guard the migration cannot run under the test database, and the whole suite fails at setup rather than at any individual assertion.

"N failed, 0 assertions" means the setup failed, not the tests. Recognising that distinction shortened a debugging session that had already consumed most of an evening.

A hard-coded configuration value defeats the test environment silently. The Product suite was broken for three weeks with no visible signal at all. When environment overrides appear not to work, read the configuration files before anything else.

When adding a cap, guard the update paths, not only creation. The adversarial review of the seat cap found six bypasses and every one was an update path.

Add the cast in the same commit as the column. Three missing datetime casts produced a live fault when a comparison ran against a string value.

A lookup on a globally-unique column must ignore company scoping. A scoped lookup misses rows belonging to other companies, and the resulting insert fails on the unique index as a server error rather than as the conflict it actually is.

A session note written before the session ends is a snapshot, not a verdict. The push status in this report is the case in point. Repository state should be verified from the repository at the start of the next session, not carried forward from a note.

*Prepared for Ejaz Hussain, Managing Director, M/s. Ametecs India Private Limited, Hyderabad.
Sources: SmartEPT project memory, sessions of 11–14 August 2026; direct inspection of both repositories, 15 August 2026.*